

Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Fall 2018

Program: B.Sc. in CSE (Batch: 7th) 1st Year 1st Semester

Course Code: PHY 1101

Course Title: Physics – I

Time: 1 hour & 30 minutes

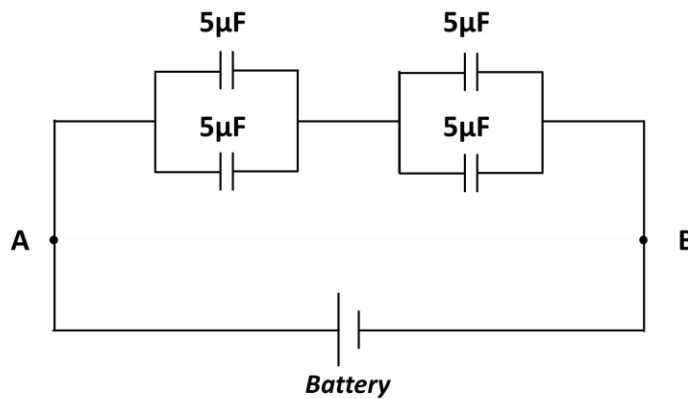
Total Marks: 30

Group-A

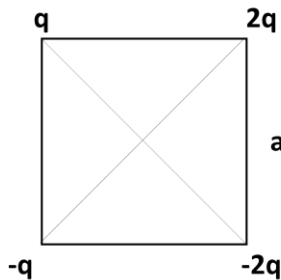
[Answer any 5 (Five) questions including No. 1.]

5 × 5 = 25

1. See the circuit diagram carefully and answer the questions that follows –



- a) Define 1 Farad. 1
- b) Calculate the equivalent capacitance of above circuit. 3
- c) How do you increase the equivalent capacitance of the above circuit arrangement? [You can give answer with diagram] 1
2. Figure shows the system of charges in a square shape of arms of a length. Answer the following questions –



Assume that, $q = 1.0 \times 10^{-8}\text{C}$ and $a = 5\text{cm}$

- a) Write down the equation for electric field for the system of charges. 1
- b) Find the electric field and electric potential at the center of the square shape. 2+2=4

3.
 - a) What is electrostatics? 1
 - b) Give some limitation of the Coulomb's law. 2
 - c) Calculate the repulsive Coulomb force exist between two protons in a nucleus of iron. Assume a separation of $4 \times 10^{-15}\text{m}$. 2

4.
 - a) State and explain Gauss's Law. 1+1=2
 - b) By applying the Gauss's law. Construct an equation for electric field due to a long-charged rod. 3

5.
 - a) Define drift velocity. 1
 - b) Define Electric field from potential gradient. 1
 - c) What is the drift velocity of electrons in a copper conductor having a cross-sectional area of $5 \times 10^{-6} \text{ m}^2$, if the current is 10A? Assume that there are 8×10^{28} electrons/ m^3 ($q_e = 1.6 \times 10^{-19}\text{C}$). 3

6.
 - a) Draw the parallel plate capacitor diagram. 1
 - b) Explain the effect of dielectric medium in the capacitor. 2
 - c) The parallel plates of an air-filled capacitor are everywhere 1.0 mm apart. What must be the plate area if the capacitance is to be 1.0 farad? 2

7.
 - a) Establish the relationship between charge density and electric field. 1
 - b) Write a short note on – : (Any two) 2+2=4
 - i) *Quantization of electric charge*
 - ii) *Equipotential surface*
 - iii) *Gaussian surface*
 - iv) *Permittivity constant & Relative permittivity*

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[Answer all questions]

0.5 × 10 = 5

Student Name:	ID No.:
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1. What is the value of the permittivity of free space?

- | | |
|---|---------------------------------|
| a) $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$ | b) $9 \times 10^9 N m^2 C^{-2}$ |
| c) $1.9 \times 10^{-19} C$ | d) $1.6 \times 10^{-19} C$ |
| e) $9.1 \times 10^{-31} kg$ | |

2. The reciprocal of resistance is called the –

- | | |
|-----------------|------------------------|
| a) resistivity | b) specific resistance |
| c) conductance | d) conductivity |
| e) permittivity | |

3. With an equality or balance of charge, the object said to be-

- | | |
|-----------------------|-------------------------|
| a) maximum charged | b) negatively charged |
| c) positively charged | d) electrically neutral |
| e) mixed charged | |

4. According to the Gauss's law, $\oint_s \vec{E} \cdot \vec{da} = ?$

- | | |
|-------------------|---------------------------|
| a) q | b) $\frac{q}{\epsilon_0}$ |
| c) $\epsilon_0 q$ | d) $\frac{\epsilon_0}{q}$ |
| e) ϵ_0 | |

5. The potential differences between two points in an equipotential surface is -

- | | |
|------|------|
| a) 0 | b) 1 |
| c) 2 | d) 4 |
| e) 8 | |

6. The stored energy of a capacitor is –

a) $w = \frac{1}{2} CV$

b) $w = \frac{1}{2} CV^2$

c) $w = \frac{1}{2} VQ$

d) $w = \frac{1}{2} CQ^2$

e) $w = \frac{1}{2} VC^2$

7. What is the unit of the relative permittivity?

a) *Volt*

b) *Joule*

c) *None*

d) *C*

e) *F*

8. The net flux of a cylindrical Gaussian surface placed in an external field is

a) negative

b) positive

c) infinity

d) Maximum

e) Zero

9. Which one is possible?

a) $q = 45e$

b) $q = 35.1e$

c) $q = 25.5e$

d) $q = 15.9e$

e) $q = 5.5e$

10. Conductance is to the cross-sectional area of the conductor.

a) equal

b) proportional

c) inversely proportional

d) independent

e) reciprocal

Invigilator Signature

Examiner Signature